

PRIMAcquisition

PRIMAcquisition (PRIM) is a Python-based application for synchronized acquisition of pressure data (from an Arduino-controlled pressure transducer) and live camera imaging. Designed for vascular physiology experiments, PRIMAcquisition displays live pressure traces, previews a high-speed camera feed, and saves synchronized recordings (as a CSV and TIFF stack) for offline analysis.

Quick Start – Run an Experiment

- 1 **Connect PRIM Device** – Select the Arduino COM port and click **Connect PRIM Device**
- 2 **Set Up Camera** – Choose camera & resolution → click **Start Camera**
- 3 **Adjust Exposure/Gain** – Use controls to fine-tune camera settings
- 4 **Zero PRIM** – Make sure pressure is at zero
- 5 **Start Recording** – Click **Start Recording** to begin acquisition
- 6 **Stop Recording** – Click **Stop Recording** when finished
- 7 **Playback & Export** – Click **Playback** to review, export a frame (PNG) or full TIFF with overlay

Features

Real-Time Pressure + Video Recording

- **Arduino is the master clock:**
 - Generates precise trigger pulses (`CamTrig`) to the camera for each frame.
 - Immediately sends synchronized serial data: `frame_index`, `elapsed_time_s`, `pressure_value`.
- **App listens for first Arduino tick**, then begins writing data.
- **Perfect sync** is maintained:
 - Each TIFF frame corresponds to exactly one Arduino trigger.

- Each pressure value corresponds to exactly one recorded frame.
- The matching pressure value is stored in each TIFF frame's metadata.

Live Pressure Plotting

- Receives pressure readings from Arduino (115200 baud).
- Displays live pressure trace in real time.
- Shows frame index, elapsed time, and pressure in the top panel.

High-Speed Camera Preview & Control

- Integrates with The Imaging Source DMK cameras via IC Imaging Control 4 (IC4).
- Lists connected USB3 Vision cameras with supported resolutions.
- Displays a full-resolution preview using OpenGL viewfinder.
- Camera settings (exposure, gain, brightness) adjustable via sliders.

Synchronized Recording Output

- Folder structure:

```
PRIM_ROOT/YYYY-MM-DD/FillN/  
├ recording_*.csv      ← Pressure + frame index  
└ recording_*.tif      ← Grayscale video, one frame per Arduino trigger
```

- Files are automatically named and saved in time-stamped subfolders.
- Environment variable `PRIM_RESULTS_DIR` overrides default save location.
- After recording you can open **Playback** to view the TIFF with pressure overlay (toggleable), export an annotated copy, or save a single-frame snapshot.

Under the Hood

PRIMAcquisition uses a **hardware-triggered acquisition model**:

Component	Role
Arduino	Master clock; sends trigger pulses and serial data
Camera	Triggered CamTrig pin from Arduino
App	Waits for first Arduino message, then saves video + CSV

Each cycle: 1. Arduino triggers a camera frame via digital HIGH → LOW on CamTrig . 2. Arduino immediately sends pressure data and timestamp via serial. 3. App receives both the frame and serial line, saves them together.

Result: **pixel-perfect and time-accurate alignment** of pressure + video frames.

- **SerialThread** – Reads Arduino serial data and emits (frameIndex, timestamp, pressure) .
- **SDKCameraThread** – Opens IC4 camera, configures settings, streams frames to the GUI.
- **RecordingManager** – Writes synchronized CSV and TIFF files in a background thread.
- **PlaybackWindow** – Reloads recorded files and overlays pressure data on frames.

Recording starts when the first Arduino tick is received. Each pressure value is embedded in its corresponding TIFF frame metadata, ensuring **perfect synchronization**.

Installation

1. Clone the Repository

```
git clone https://github.com/your-repo/PRIMAcquisition.git
cd PRIMAcquisition/prim_app
```

```

### 2. **\*\*Create a Virtual Environment\*\***

```
```bash
python -m venv .venv
```

```
.venv\Scripts\activate # On Windows
```

2. Install Dependencies

```
pip install -r requirements.txt
```

Or manually:

```
pip install pyqt5 imagingcontrol4 pyserial numpy tifffile
```

3. Install IC4 SDK

- Required for DMK camera support.
- Ensure GenTL Producer is installed and camera enumerates via IC4.

Running the App

1. Start App

```
python prim_app.py
```

2. Connect Arduino

- Select COM port (e.g., COM8), then click **Connect PRIM Device**.
- Confirm status shows “Connected”.

3. Configure Camera

- Select camera and resolution.
- Click **Start Camera** to preview live feed.

4. Start Recording

- Use menu: **Acquisition** → **Start Recording** or press **Ctrl+R**.
- App waits for first Arduino 'T...' tick to begin synchronized recording.
- TIFF and CSV are saved in the correct folder.

Stop Recording

5.

- Use menu: **Acquisition** → **Stop Recording** or press **Ctrl+T**.
- Files are finalized and closed cleanly.

Arduino Firmware (PRIMv302)

- Loops every `startup.timeDelay` milliseconds.
- Sends a `CamTrig` HIGH–LOW pulse to trigger one camera frame.
- Immediately sends formatted serial data:

```
frame_index, elapsed_time_s, pressure_value
```

- Operates with precise timing via `micros()` and `millis()`.

Troubleshooting

Issue	Fix
Camera not listed	Verify IC4 SDK + GenTL Producer installed
Serial shows no data	Check Arduino COM port and baud rate
TIFF won't open	Use ImageJ/Fiji or Python <code>tifffile</code>
Dropped frames	Use USB 3.0 and reduce resolution if needed

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